

Characterizing Risks from Propulsion System Failures in Terminal Operations in Urban Air Mobility Settings

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Abstract

Urban Air Mobility (UAM) comes with the addition of many small eVTOL aircraft into congested, low-altitude airspaces, bringing new safety and regulatory issues to light. Unlike traditional commercial aviation, UAM operations result in larger quantities of small aircraft flying much closer to where people live and work. If regulations are too strict, it could deter investment into UAM, but if regulations are too loose, it could raise the risk for passengers and bystanders. This project aims to study the challenges associated with eVTOL propulsion systems and identify the key risks for UAM operations. We will conclude with making suggestions on methods to make UAM operations safer and easier to manage.

Nomenclature

AAM – Advanced Air Mobility

ATC – Air Traffic Control

BMS – Battery Management System

CONOPs – Concept of Operations

DEP – Distributed Electric Propulsion

EMI – Electromagnetic Interference

EPU – Electric Propulsion Unit

ESC – Electronic Speed Controller

eVTOL – Electric Vertical Takeoff and Landing

FAA – Federal Aviation Administration

FCC – Flight Control Computer

FMEA – Failure Modes and Effects Analysis

FTA – Fault Tree Analysis

HV – High Voltage

IGE – In Ground Effect

L+C – Lift + Cruise

NASA – National Aeronautics and Space Administration

PMSM – Permanent Magnet Synchronous Motor

RPM – Revolutions Per Minute

OGE – Out of Ground Effect

SoC – State of Charge

STPA – System-Theoretic Process Analysis

UAM – Urban Air Mobility

UCA – Unsafe Control Action

UOE – UAM Operations Environment

VRS – Vortex Ring State

1 Introduction

Urban Air Mobility (UAM) is quickly advancing as a potential solution to urban congestion by using electric vertical takeoff and landing (eVTOL) aircraft capable of moving people and goods at low altitudes above cities. Under the broader Advanced Air Mobility (AAM)¹ framework, NASA and the Federal Aviation Administration (FAA) outlined concepts of operations (ConOps) that envision a system for the future designed around dense traffic, rooftop and ground-level vertiports, and large fleets of small aircraft operating near people on the ground (FAA, 2024). These documents emphasize that UAM operations eventually must meet commercial aviation-

¹ Advanced Air Mobility (AAM) is the broader category of new electric and automated air transport systems. UAM is the urban subset focused on short-range city operations.

level safety while relying on fundamentally different technologies, such as distributed electric propulsion (DEP), high-energy batteries, and highly automated flight control and traffic-management systems (Fontaine, 2023).

The FAA’s UAM ConOps outlines the basic operational framework for this system, describing air routes designed to accommodate many simultaneous eVTOL flights while linking a network of dispersed vertiports and interfaces and integrating with existing controlled airspace (FAA, 2024). Unlike traditional aviation centered on large airports, UAM aircraft will operate at low altitudes, with many more aircraft flying closer to people, buildings, and other obstacles, and with irregular traffic patterns driven by on-demand operations (Hill et al., 2020). Figure 1 below shows a representative UAM Operating Environment (UOE), showcasing the density and complexity being anticipated for mature UAM operations.

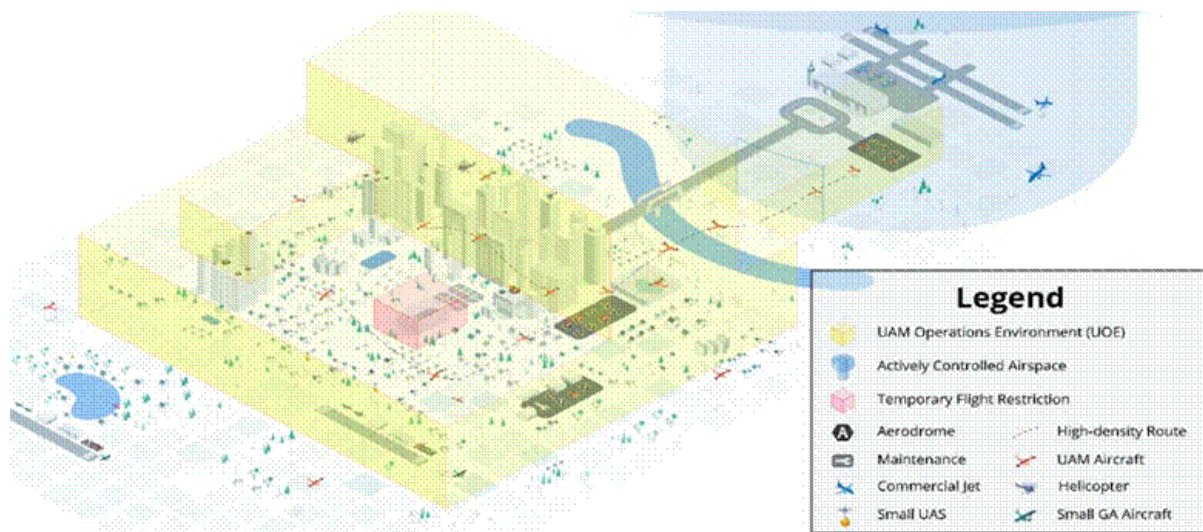


Figure 1: Isometric Operational View of a Representative UOE (Hill et al., 2020)

Ensuring safety in this new operational environment is one of the largest barriers to widespread UAM adoption. NASA’s UAM Vision ConOps describes challenges spanning from ground infrastructure requirements to airspace management, automation, and overall system integration (Hill et al., 2020). The central underlying topic in all of these issues is the need to establish the airworthiness and reliability of the eVTOL aircraft themselves. This would rely heavily on new electric propulsion technologies like DEP, a key enabler of the eVTOL performance and control. DEP offers improved control and can improve redundancy and maneuverability, but it also

introduces new complexities in power distribution, thermal management, software control, and component reliability (Kim et al., 2020).

Many eVTOL designs cannot perform safe autorotation like conventional rotorcraft; this is because of small rotor disks and high disk loading, making propulsion system integrity very important throughout the entire flight envelope (eVTOL.news, 2024). Accidents like the 2022 Joby JAS4-2 rotor blade failure show how structural or control failures in a DEP system can cascade quickly and result in catastrophic outcomes (Vertical Magazine, 2022). Other existing literature provides other small bits of relevant insight. NASA and American Institute of Aeronautics and Astronautics studies examined mission requirements, vertiport design, and initial safety analyses using STPA or functional hazard assessments (Basset et al., 2018). Several component level studies also investigated motor degradation, thermal runaway in batteries, and electronic speed controller (ESC) failure dynamics (Belharouak, 2024).

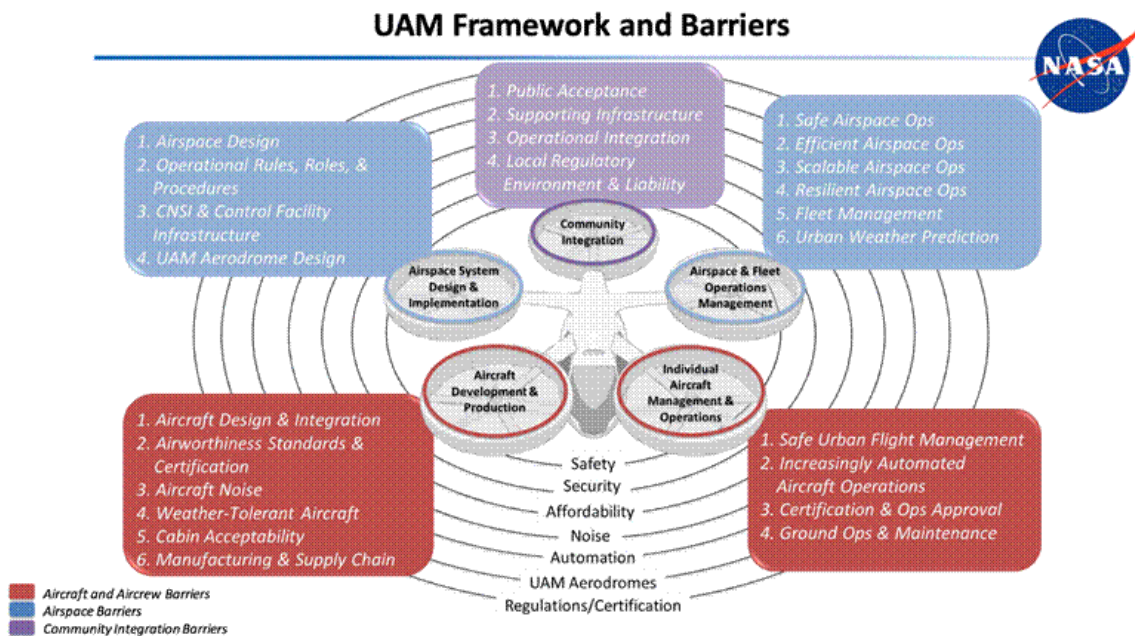


Figure 2: UAM Framework and Barriers (Hill et al., 2020)

However, scholars and regulators still have not properly characterized how propulsion system failures behave during the most safety-critical phases of UAM operations like takeoff, out of ground effect hover, transition, approach, and vertical landing, where power demand is highest and margins for recovery are small.

This gap is urgent because:

- UAM aircraft fly very close to people and structures, leaving almost no energy or altitude margin for recovery from propulsion failures.
- DEP systems are susceptible to common-mode and cascading failures, and NASA simulations show that these failure chains can unfold in under 250 ms (NASA, 2021).
- Battery and thermal loads are highest during terminal operations. This increases the likelihood of power limitations or component degradation during flight (Dixit, 2024).
- UAM safety targets are around 10^{-9} fatal accidents per flight hour. This is similar to commercial aviation but with completely different technology (NASA, 2021).

This paper addresses this deficiency by developing a structured framework to characterize propulsion system failures across eVTOL architectures and identifies the risk associated with specific flight-phases. Using FMECA, Fault Tree Analysis, and STPA, we analyze failure modes in motors, batteries, and power distribution hardware, and explain how they evolve into collective or cascading events during terminal operations. We then link these risks to corresponding hardware, software, operational, and certification mitigations, with the broader goal of informing future safety cases for UAM.

To structure our work, we defined four guiding research questions.

1. First, we ask: What are the key propulsion system failure modes in eVTOLs, specifically electric motors, batteries, and power electronics, and how do they differ from the failure modes we're used to in conventional aviation?
2. Second, because eVTOLs often use distributed electric propulsion, we ask: How do single failures combine into common-mode and cascading events? For example, can the loss of one motor overload the others and trigger a chain reaction?
3. Third, since we are narrowing our scope on the terminal operations of vertical takeoff, hover, transition, and landing, we ask: Which phases are most sensitive to these failures and why are they harder to manage than comparable phases in traditional aircraft?

4. Finally, we explore mitigation options: Which combinations of hardware design, fault-tolerant control, operational procedures, and certification requirements are most effective in bringing propulsion-related risk down to an acceptable level?

Our system analysis on risks in eVTOL operations starts with the System-Theoretic Process Analysis (STPA) to look at the “Bigger Picture.” We then drill down into the specific failure modes of the hardware using risk matrices to map our findings and prioritize mitigation.

To keep the project tractable, we bound our scope to the propulsion and power system, assuming that other subsystems, such as communications and air traffic management, are operating nominally.

2 Risks in UAM Operations

2.1 Propulsion System Architectures

Our analysis will focus exclusively on all-electric configurations centered on the Distributed Propulsion System (DEP). DEP is a propulsion concept where multiple electric propulsors are integrated across the airframe, powered by an electrical power system rather than mechanical driveshafts (Kim et al., 2020). This approach offers significant benefits in terms of redundancy, noise reduction, and advanced flight control, but it also introduces new complexities in power distribution, thermal management, and component reliability (Silva, 2023).

2.2 DEP Architectures

For this analysis, DEP architectures are grouped into three primary categories, originating from categories defined by NASA (See Figure 3):

- **Multirotor:** Vehicles relying exclusively on rotor speed for lift and control (e.g., quadcopter, octocopters, etc.). While mechanically simple, they depend heavily on ESC reliability; common-mode electrical failures can negate rotor redundancy. They have VTOL capabilities without the transition phase; however, the small rotor size generally precludes effective autorotation.
- **Lift + Cruise (L+C):** Designs using independent rotors for vertical lift and forward flight (e.g., Joby S4). This separation offers redundancy but increases component count and drag with its increased mechanical complexity involving tilting rotors to transition from lift to

cruise. A failure of the cruise propeller can critically compromise both forward thrust and pitch authority.

- **Vectored Thrust:** Aircraft that tilt rotors or wings to transition between flight modes (e.g., NASA GL-10). This maximizes efficiency but introduces single points of failure in the mechanical tilting actuators, making the aerodynamic transition phase particularly vulnerable to loss of control.



Figure 3: NASA concept eVTOL aircraft. Clockwise from Top-Left are Vectored Thrust, Multirotor, and L + C

2.3 Propulsion Risks Across Flight Phases

The risk profile of the electric propulsion system changes dramatically with the demands of each flight phase. The 3 key flight phases are distinctly illustrated in Figure 4. A comprehensive safety assessment must analyze the unique vulnerabilities present in each of these phases, a topic which will be explored further in section 4.4.

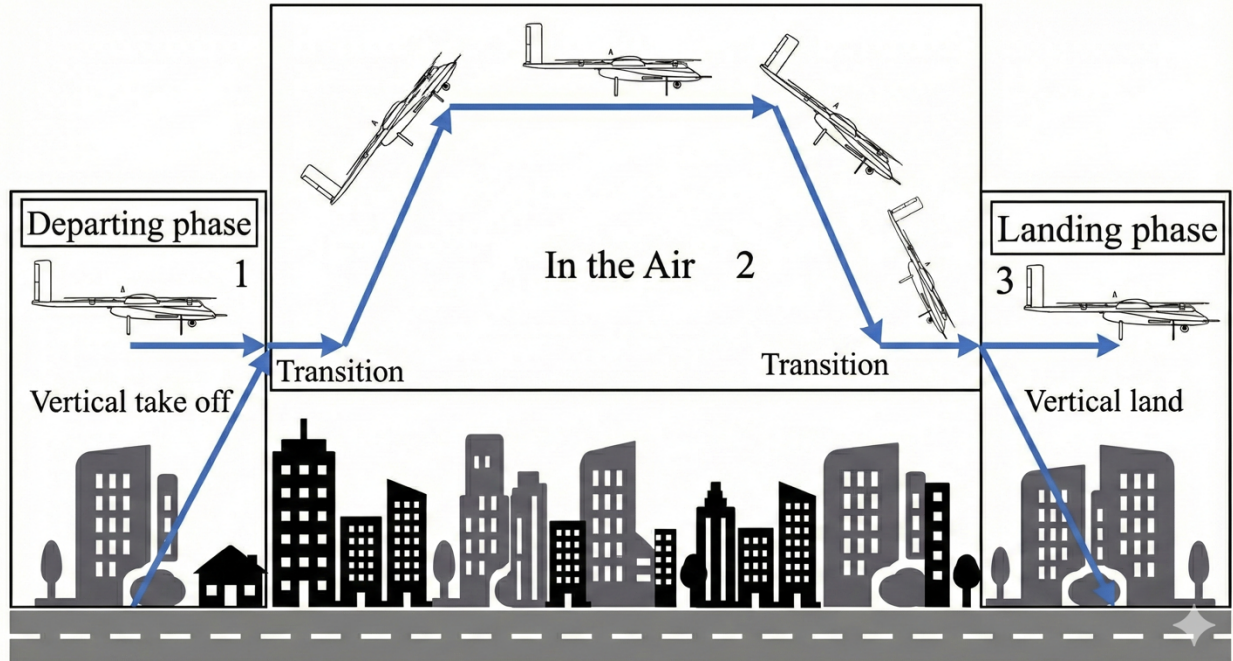


Figure 4: Flight Phases for UAM aircraft

- Takeoff and Landing:** These terminal operations are the most power-intensive phases of flight, requiring sustained high-power output to generate vertical lift. This places maximum thermal and electrical stress on all propulsion components. Operations occur at low altitudes and low air speeds, providing minimal time or energy to recover from failure. Landing requires high-precision control to navigate into a constrained vertiport environment, which may be surrounded by obstacles such as buildings. For example, these regions can introduce hazards like the vortex ring state (VRS), which can lead to a sudden loss of lift and control during a descent (Amo et al., 2022).
- Lift (Climb/Transition):** For vectored thrust and L+C designs, the transition from vertical, rotor-borne lift to horizontal; wing-borne flight is a critical phase. During this maneuver, the aircraft is in a complex aerodynamic state, often at high angles of attack and with changing thrust vectors (May et al., 2025). A failure of a lift rotor or a tilting mechanism during this phase can lead to a sudden loss of altitude or control before the aircraft has achieved sufficient forward airspeed for its wings to generate adequate lift.
- Cruise:** While typically the most energy-efficient phase, cruise presents endurance-related risks. Prolonged operation can expose weaknesses in thermal management systems, leading to the overheating of motors or batteries. This phase also increases exposure to

environmental hazards like icing or bird strikes. Critically, the cruise phase depletes the battery's state of charge, which reduces its ability to deliver peak power. This is a significant concern, as sufficient power must be reserved for the high demands of the subsequent landing phase (Belharouak, 2024; Dixit, 2024).

- **Descent:** The descent phase requires precise energy management to ensure sufficient power reserves are available for the final approach and a potential go-around or balked landing. A propulsion system failure during the final vertical descent, when the aircraft is at low altitude and has low potential energy, could be catastrophic.

3 System-Theoretic Process Analysis (STPA)

To effectively characterize the risks in Urban Air Mobility, we adopted **STPA**, a hazard analysis technique detailed in the *STPA Handbook* (Leveson & Thomas, 2018). Unlike traditional safety assessment methods that treat safety as a reliability problem, focusing on preventing individual component failures, STPA treats safety as a dynamic control problem. This distinction is important for eVTOL architecture since DEPs are tightly coupled complex systems. Accidents can occur even when all subsystems function exactly as specified; this is often due to unsafe interactions or conflicting requirements between controllers (e.g., if a Battery Management System (BMS) correctly shuts down a battery to prevent a fire, inadvertently causing a crash due to loss of thrust). By applying STPA, we analyze the system top-down to identify such interactions and logic flaws before diving into the component specifics.

3.1 Step 1: Operational Control Structure

Risk in terminal operations shifts as the aircraft moves through different control regimes. We analyzed the hierarchical control structure across three distinct operational phases derived from the flight envelope shown in Figure 4: Phase 1 (Departing), Phase 2 (In the Air), and Phase 3 (Landing). The overarching operational control structures for the three flight phases have been adapted from the paper titled “Safety Analysis of eVTOL Operations based on STPA” (Elizabeth et al., 2025).

3.1.1 Phase 1

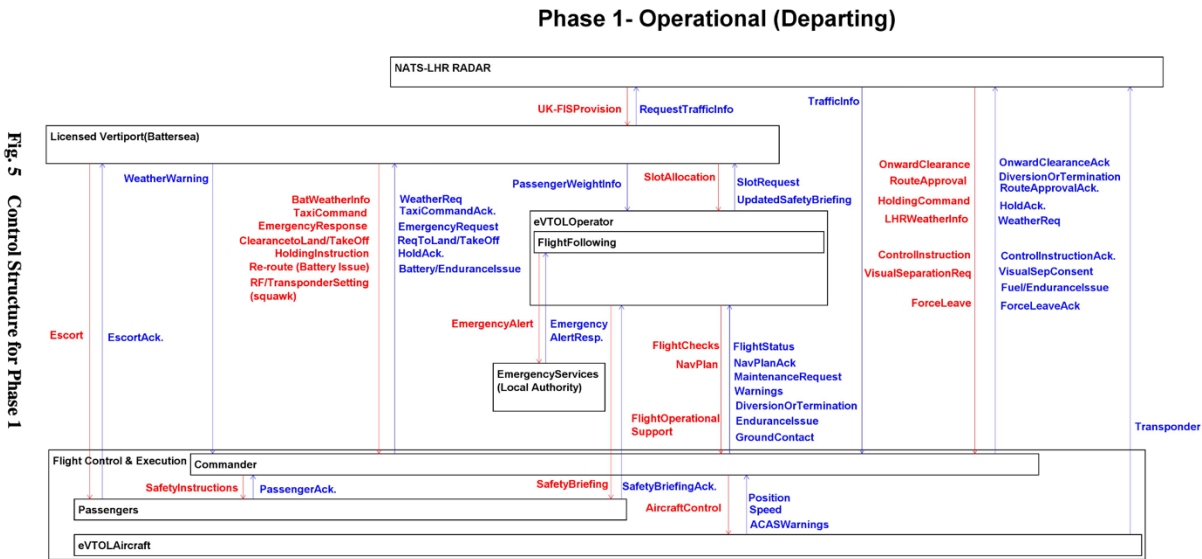


Figure 5: Departing Phase Operational Control Structure (Elizabeth et al., 2025)

This phase encompasses the vertical takeoff sequence and initial climb. As shown in Figure 5, the control structure during departure involves coordination between the Licensed Vertiport, eVTOL Operator, and the aircraft itself. The vertiport issues clearances and monitors battery/weather conditions, while the operator manages passenger loading and flight following. From a propulsion perspective, this phase demands peak power output for vertical lift, placing maximum stress on the battery system and motor drives. The control structure must handle taxi commands, takeoff clearances, and emergency response coordination in the event of a propulsion anomaly during this power-critical phase.

3.1.2 Phase 2

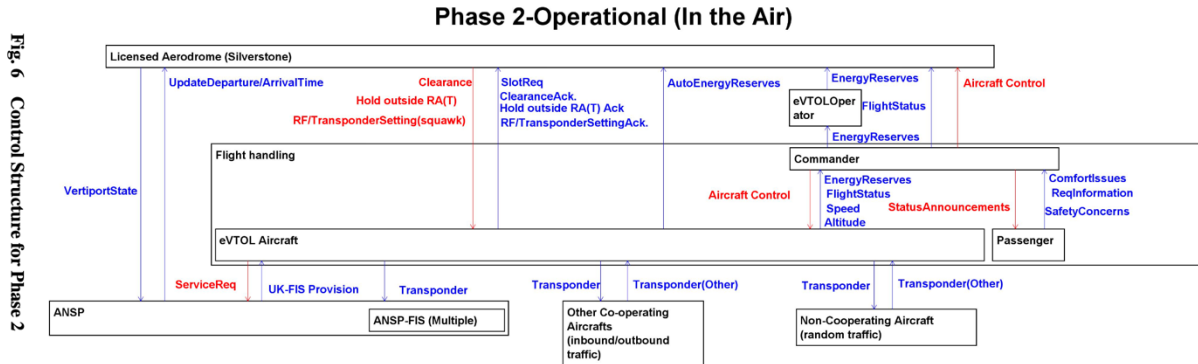


Figure 6: In the Air Phase Operational Control Structure (Elizabeth et al., 2025)

Once airborne and in cruise, the control structure shifts to emphasize airspace coordination, as depicted in Figure 6. The Licensed Aerodrome and Air Navigation Service Providers (ANSPs) become the primary external controllers, managing slot allocations, flight status updates, and energy reserve monitoring. For the propulsion system, this phase involves sustained but moderate power demands, making thermal management and state-of-charge monitoring the primary concerns. The "eVTOL Aircraft" box must continuously report energy reserves and flight status while responding to hold or diversion commands that may affect propulsion system planning.

3.1.3 Phase 3

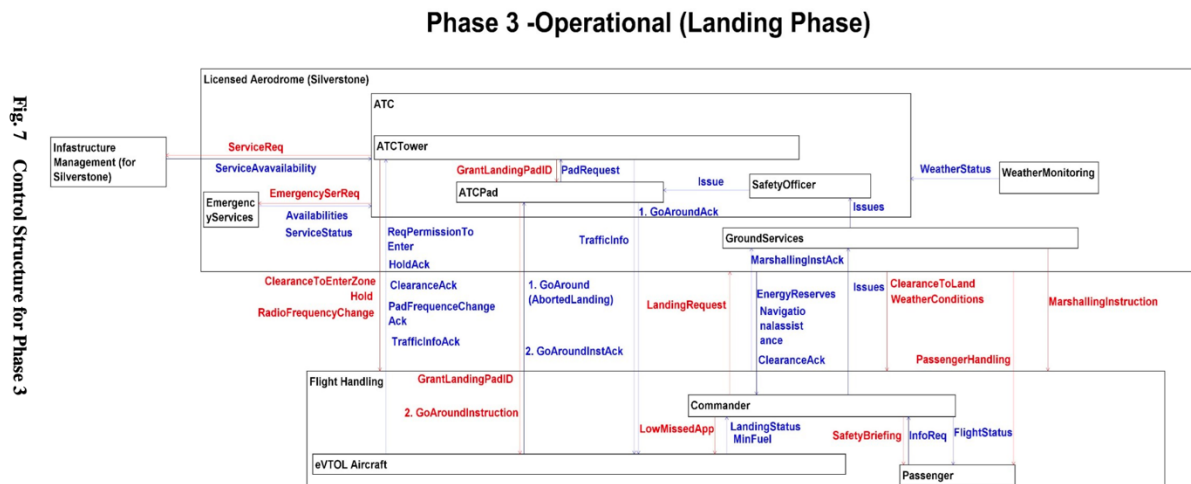


Figure 7: Landing Phase Operational Control Structure (Elizabeth et al., 2025)

The final approach and vertical landing phase, shown in Figure 7, concentrates multiple operational risks into a narrow time and altitude window. The control structure involves the Licensed Aerodrome, ATC, Ground Services, and Safety Officers coordinating landing pad assignments, go-around decisions, and marshalling instructions. For the propulsion system, this phase is particularly critical: power demand peaks again for hover and descent arrest, often when battery state-of-charge is at its lowest. The control structure must support rapid go-around commands, low-energy warnings, and emergency landing coordination—all while the propulsion system operates near its thermal and electrical limits.

3.1.4 Unacceptable Losses

Within these operational control structures, the "eVTOL Aircraft" box represents the physical vehicle and its onboard systems. Since our analysis specifically targets propulsion-related hazards, we expand this box to reveal the internal control architecture of the propulsion system. Figure 8 in Section 3.3 presents this detailed propulsion system control structure, showing the hierarchical relationships between subsystems and the bidirectional flow of control commands and feedback signals. However, before going into that subsystem, using the control structures from the 3 phases above, we identified the unacceptable losses and the system-level hazards that must be prevented.

Table 1: Defined Unacceptable Losses for eVTOL Operations

Loss ID	Loss Name	Description
L-1	Loss of Life or Injury	Death or serious injury to passengers, crew, ground personnel, or the general public.
L-2	Asset Damage / Hull Loss	Significant damage to or total loss of the eVTOL aircraft, ground infrastructure (vertiports), or third-party property.
L-3	Loss of Mission / Service	Inability to complete the transport mission, leading to diversion, delays, or unavailability of the aircraft.
L-4	Environmental Contamination	Release of toxic substances (e.g., battery electrolytes) or generation of hazardous debris (e.g., rotor fragments).
L-5	Loss of Public Confidence	Damage to the reputation of the operator or the UAM industry, potentially leading to regulatory shutdowns.

3.2 Step 2: System and subsystem-level hazards

Using the control structures defined in Section 3.1, we identified the unacceptable losses and the system-level hazards that must be prevented. This follows the standard STPA methodology where losses represent the high-level outcomes that the system must avoid, and hazards are system states or conditions that, in combination with environmental factors, could lead to those losses.

Table 1 presents the five categories of unacceptable losses identified for UAM propulsion systems. These losses range from immediate safety concerns (L-1: Loss of Life or Injury) to longer-term consequences for the industry (L-5: Loss of Public Confidence). The inclusion of L-4 (Environmental Contamination) and L-5 reflects the unique context of UAM operations in densely populated urban areas, where even non-fatal incidents could have significant environmental and reputational consequences.

Table 2: System-Level and Propulsion Hazards with Linked Losses

Hazard ID	System-Level Hazard (H)	Propulsion-Specific Sub-Hazards	Linked Losses
H-1	"Aircraft unable to maintain controlled flight or attitude (Condition: Thrust/Lift < Weight or Moments > Control Authority)"	H-1.1: Total Loss of Thrust: Simultaneous failure of all Electric Propulsion Units (EPUs) leading to inability to maintain altitude.	L-1, L-2, L-5
		H-1.2: Asymmetric Thrust: Differential thrust between motors exceeds flight control authority, causing uncommanded roll/yaw.	
		H-1.3: Transition Failure: Propulsion system fails to rotate/tilt mechanisms, trapping aircraft in vertical or cruise mode.	
		H-1.4: Slow Thrust Response: Motor spin-up/spin-down latency is too slow to stabilize aircraft in high-gust environments.	

H-2	"Aircraft experiences uncontrolled release of hazardous energy (Condition: Heat, Voltage, or Kinetic Energy escapes containment)"	H-2.1: Thermal Runaway Propagation: Failure of BMS to contain a single-cell fire, leading to pack-level fire or explosion.	L-1, L-2, L-4
		H-2.2: High Voltage (HV) Arc: Breakdown of insulation in distribution bus causes arcing to fuselage or critical avionics.	
		H-2.3: Rotor/Blade Liberation: Structural failure of a propeller blade releases high-energy debris towards the fuselage.	
		H-2.4: Toxic Gas Release: Battery venting releases toxic electrolytic gases into the passenger cabin or ground environment.	
H-3	"Aircraft operates with insufficient energy reserves (Condition: Energy Available < Energy Required to Land)"	H-3.1: Voltage Collapse: Battery voltage sags below inverter cutoff threshold during high-power demand (e.g., go-around).	L-1, L-2, L-3
		H-3.2: Inaccurate State of Charge (SoC) Estimation: BMS reports higher energy available than exists, causing exhaustion before landing.	
		H-3.3: Excessive Consumption: Flight control logic "fights" motors (e.g., yaw vs. roll commands), draining battery faster than predicted.	
H-4	"Aircraft generates hazardous surface movement or forces (Condition: Unintended thrust	H-4.1: Uncommanded Spin-Up: Motors activate on the ground while passengers or ground crew are near rotors.	L-1, L-2
		H-4.2: Braking Failure: Electric motors fail to provide regenerative braking torque during taxi or landing roll.	

	on ground or lack of braking)"	H-4.3: Inadvertent Reverse: Reverse thrust is activated airborne or asymmetrically on the ground.	
H-5	"Aircraft exceeds operating envelope (Condition: System pushed beyond thermal/structural limits)"	H-5.1: Inverter Overheat: Power electronics exceed thermal limits, triggering a protective shutdown during critical flight phase.	L-2, L-3
		H-5.2: Bus Over-Voltage: Regenerative braking generates voltage spikes exceeding the DC bus or battery limits.	
		H-5.3: Propeller Overspeed: Motor RPM exceeds structural limits of the propeller blades (e.g., during a dive).	

The hazard decomposition in Table 2 reveals several important characteristics of propulsion system risk. H-1 (Loss of Controlled Flight) encompasses the most direct pathway to catastrophic outcomes, spanning both abrupt failures such as total loss of thrust and more gradual degradations like slow thrust response. H-1.2 (Asymmetric Thrust) indicates a failure mode unique to DEP architectures where differential thrust between motors can exceed the flight control system's authority. H-2 (Uncontrolled Energy Release) addresses the high energy density inherent in eVTOL propulsion systems, from battery thermal runaway to blade liberation events. H-3 (Insufficient Energy Reserves) captures hazards that may not manifest as immediate failures but can lead to catastrophic outcomes during demanding flight phases, particularly during the landing phase when reserves are lowest. H-4 (Hazardous Ground Operations) addresses risks specific to vertiport environments, while H-5 (Operating Envelope Exceedance) recognizes that propulsion components have thermal and structural limits that, if exceeded, can trigger protective shutdowns at critical moments.

3.3 Propulsion Sub-Hazards

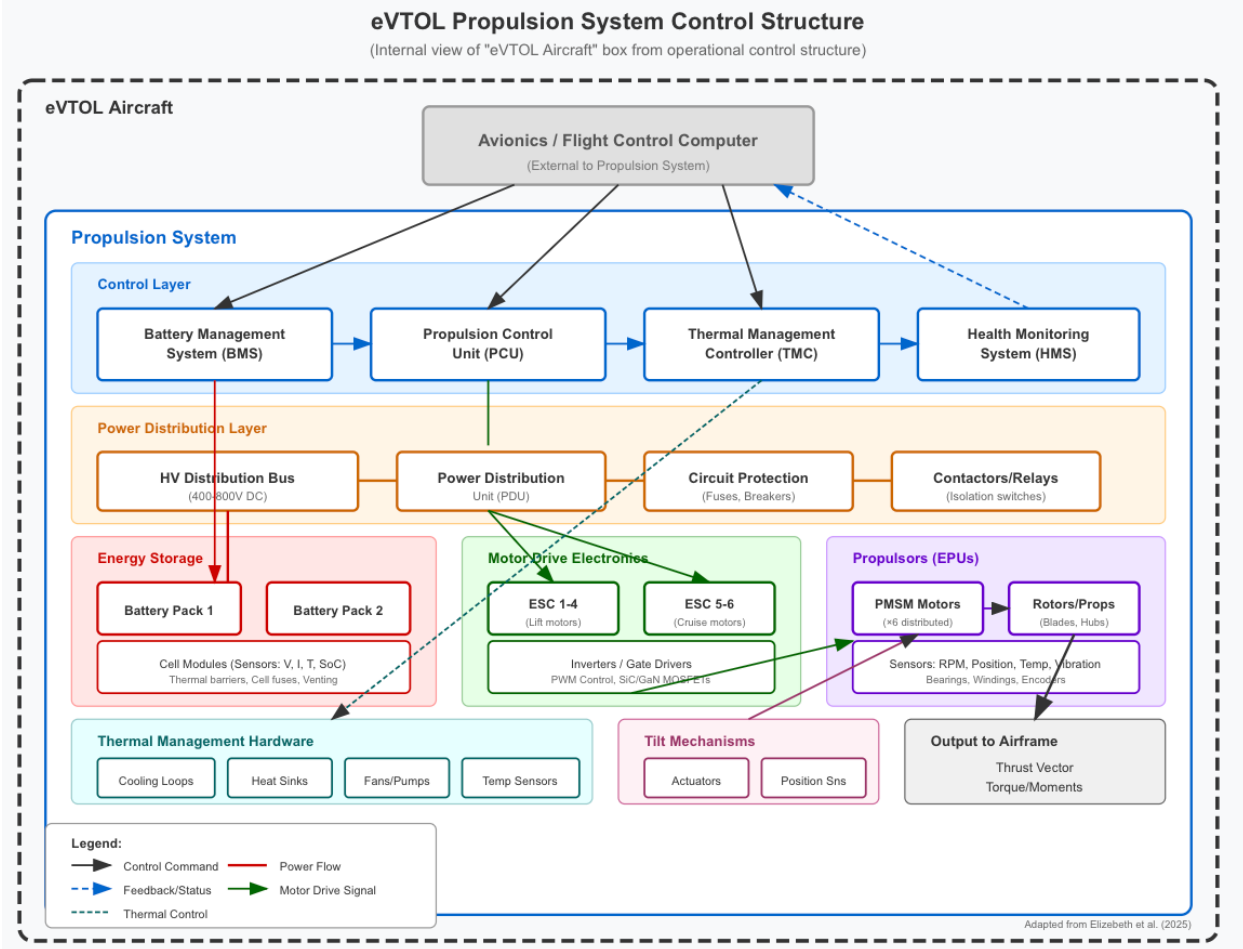


Figure 8: Propulsion System Control Structure

The propulsion system control structure, shown in Figure XX, is organized into several functional layers that interface with the Flight Control Computer (FCC) and coordinate all propulsion functions:

Control Layer: At the top of the propulsion hierarchy, this layer includes the Battery Management System (BMS), which monitors cell-level parameters and enforces safe operating limits; the Propulsion Control Unit (PCU), which translates flight control commands into motor speed and torque setpoints; the Thermal Management Controller (TMC), which regulates cooling system operation; and the Health Monitoring System (HMS), which aggregates sensor data for fault detection and prognostic warnings.

Power Distribution Layer: This layer manages the flow of electrical energy from storage to propulsors. The High-Voltage Distribution Bus operates at 400–800V DC and serves as the power backbone. The Power Distribution Unit (PDU) routes power to individual motor channels, while circuit protection devices isolate faults before they propagate.

Energy Storage Layer: The battery system comprises multiple packs for redundancy, each containing cell modules with integrated sensors. Safety features include cell-level fuses, thermal barriers, and venting systems.

Motor Drive Electronics Layer: Electronic Speed Controllers (ESCs) interface between the DC power bus and the AC motors, performing high frequency switching to generate precise waveforms for motor control.

Propulsor Layer: The EPU consists of Permanent Magnet Synchronous Motors (PMSMs) and the rotor/propeller assemblies. Embedded sensors measure RPM, position, temperature, and vibration.

Auxiliary Systems: Supporting the primary propulsion chain are the thermal management hardware and, for vectored thrust configurations, the tilt mechanisms with their actuators and position sensors.

The arrows in Figure 8 indicate the direction of control authority (downward) and feedback/status information (upward). A critical observation from this structure is the presence of multiple feedback loops and cross-couplings that create potential pathways for unsafe control actions.

3.4 Step 3: Unsafe Control Actions (UCAs)

An observation from the structure in Figure 8 is the presence of multiple feedback loops and cross-couplings that create potential pathways for unsafe control actions. We now analyze Unsafe Control Actions (UCAs) specific to the propulsion system. Following the STPA methodology, UCAs are control actions that, in a particular context and worst-case environment, can lead to a hazard. For each controller in the propulsion system hierarchy, we systematically examine some types of potentially unsafe control actions in Table 3:

Table 3: Unsafe Control Actions

ID	Controller	Control Action	Unsafe Control Action (Context)	Hazard Link
UCA-1	BMS	Open Main Contactor	BMS opens main contactors during takeoff or landing due to a transient current spike or sensor noise (prioritizing battery protection over flight).	H-1, H-1.1
UCA-2	BMS	Open Main Contactor	BMS fails to open main contactors after a crash impact, leaving high-voltage buses energized.	H-2, H-2.2
UCA-3	BMS	Limit Current	BMS provides a restrictive discharge current limit too late during a Go-Around maneuver, causing voltage to sag below inverter cutoff.	H-1, H-3.1
UCA-4	FCC	Command Thrust	FCC commands maximum thrust when battery SoC is critical or temperature is high, triggering a BMS protective disconnect.	H-1, H-5
UCA-5	FCC	Command Thrust	FCC commands asymmetric thrust (e.g., high left, low right) when the aircraft is in level flight due to sensor fusion errors.	H-1, H-1.2
UCA-6	FCC	Command Regen	FCC commands regenerative braking (negative torque) when the battery is at 100% SoC, causing a DC bus over-voltage spike.	H-2, H-5.2
UCA-7	FCC	Reallocate Thrust	FCC fails to increase thrust on remaining motors immediately after a single motor failure is detected (cascading loss of control).	H-1, H-1.1
UCA-8	FCC	Tilt Rotors	FCC commands rotors to tilt forward (transition to cruise) too early, before wing lift is sufficient to support aircraft weight.	H-1, H-1.3

UCA-9	FCC	Tilt Rotors	FCC commands rotors to tilt upward (transition to hover) too late during approach, resulting in excessive forward speed at touchdown.	H-5, L-2
UCA-10	Pilot	Command Go-Around	Pilot commands a Go-Around when energy reserves are insufficient to re-enter the traffic pattern safely.	H-3, L-3
UCA-11	Pilot	Disable Automation	Pilot disables flight envelope protections (FCC limits) during a complex transition maneuver in turbulence.	H-1, H-5
UCA-12	BMS	Stop Charging	BMS fails to stop charging when a cell reaches thermal limits while on the ground at a vertiport.	H-2, H-2.1
UCA-13	ESC	Commutate Motor	ESC provides phase current out of sequence (loss of synchronization) during high-torque hover, causing loss of lift on that node.	H-1, H-5.1
UCA-14	FCC	Engage Autopilot	FCC engages vertical descent mode too early while still moving forward at high speed (Vortex Ring State susceptibility).	H-1, H-5
UCA-15	FCC	Spin Up Motors	FCC spins up motors while on ground when passengers or ground crew are detected within the safety exclusion zone.	H-4, H-4.1

The 15 UCAs listed in Table 3 represent some critical high-level interactions identified during this analysis. It is important to note that this list is not exhaustive. A complete, certification-grade STPA applied to a commercial eVTOL aircraft would likely derive hundreds of specific UCAs, covering almost every permutation of flight mode, environmental condition, and sub-component state. However, these selected UCAs capture the primary systemic risks inherent to DEPs and these UCA identification reveals several cross-cutting themes that inform the detailed component analysis in Section 4:

Timing criticality: Many hazardous conditions arise not from incorrect control actions per se, but from correct actions applied at the wrong time. The narrow timing windows available during terminal operations make timing-related UCAs particularly dangerous.

Mode confusion: The propulsion system must operate differently across flight phases. UCAs can arise when controllers have incorrect information about the current flight mode or when transitions between modes are not properly coordinated.

Feedback dependencies: Many UCAs become hazardous specifically because of incorrect or missing feedback. The HMS plays a critical role in cross-validating sensor readings.

Cascading effects: The tight coupling within DEP systems means that a UCA in one subsystem can propagate rapidly to others.

The UCAs identified in this section establish the framework for the detailed component-level failure mode analysis presented in Section 4, where we examine the specific hardware and software mechanisms through which these unsafe control actions can occur.

4 Key Propulsion Components and Failure Modes

This section examines the primary propulsion components and their respective failure modes. Drawing from experimental testing, industry reports, and regulatory guidance, we show how component-level degradations or abrupt failures can propagate into system-wide hazards, and we identify engineering strategies employed to mitigate these risks.

4.1 Electric Motor Failures

Electric motors serve as the core propulsion mechanism in DEP architectures used in UAMs, where multiple rotors provide both thrust and control authority. Among these, permanent magnet synchronous motors (PMSMs) are favored for their high torque density and efficiency. PMSMs operate by using stator-generated rotating magnetic fields to induce synchronous rotation in a rotor embedded with rare-earth permanent magnets, typically neodymium-based alloys. This electromagnetic interaction enables precise torque control and rapid response essential for vertical lift and hover stability (He, Q., & Wang, H., 2025). However, multiple studies, like the reliability assessments by NASA and industry safety boards, have identified recurring failure modes in eVTOL propulsion motors that present both immediate and latent risks. These include mechanical

faults such as bearing degradation, thermal and magnetic failure of rotor components, and electrical issues such as winding short circuits and phase desynchronization (Kufeld, 2021; Johnson, W., & Silva, C., 2021). This section outlines these failure mechanisms in detail and explores the engineering responses designed to mitigate their impact.

- **Bearing failures** arise from intense radial and axial loads under low-speed, high-thrust conditions that exacerbate fatigue due to degraded lubrication, shaft currents, and particulate ingestion, potentially leading to rotor imbalance or seizure (R. Darmstadt, P., 2022). These failures are insidious as they may evolve undetected unless mitigated through integrated condition-monitoring systems like accelerometer-based vibration diagnostics (R. Darmstadt, P., 2022; Benini, A., 2019), enabling predictive intervention before critical damage propagates.
- **Thermal stress** compounds mechanical wear. Permanent Magnet Synchronous Machines (PMSMs), favored in eVTOL designs for their energy density, routinely operate near thermal thresholds during hover, reaching winding temperatures of 250°C (Tallerico, T. F., 2021). At these temperatures, insulation breakdown and rotor magnet degradation become significant risks. NdFeB magnets used in PMSMs lose coercivity over time, requiring more current to deliver the same torque—an effect that imposes electrical and thermal penalties on other motors in distributed propulsion configurations (Niedra, J. M., 2001). Back-EMF estimation provides a mitigation pathway by enabling real-time detection of demagnetization (Wen, B., 2022), supporting derating or load balancing strategies.
- **Rotor and blade structural failures** are another critical vector, with empirical data from Joby Aviation showing that fatigue-induced cracking can trigger chain-reaction failures across adjacent propulsors (Hardee, H., 2024; Head, E., 2023). Embedding fiber-optic or piezoelectric sensors within blade composites supports real-time strain tracking and preemptive maintenance, a trend already adopted by over 40% of new rotorcraft platforms (Dubey, V., 2025).
- Over time, **gradual degradation** via increased resistance, friction, or eccentricity degrades overall motor efficiency and thrust output (Quiñones-Grueiro, M., 2021). NASA’s studies show that such losses convert more input power into heat rather than thrust (Tallerico, T.

F., 2021). Back-EMF monitoring again serves as a key diagnostic for early fault detection (Wen, B., 2022), making it a cornerstone of health-aware control strategies.

- In distributed electric propulsion (DEP) systems, **cascading motor failures** remain at a systemic risk. A failed unit demands up to 90% load from remaining motors (Khanouki, M. A., 2025), pushing them to their limits. Control architectures must therefore adapt dynamically—throttling bandwidth and modifying allocation logic (Su, J., 2024)—to avoid oversaturation and preserve flight stability during partial-power operation.

Table 4: Risk Matrix for Propulsion Component Failure Modes

Category	Likelihood	Impact	Risk Level
Bearing wear/ failure	Likely	Moderate	Med-High
Thermal overload	Possible	Moderate	Medium
Winding short circuits	Unlikely	Severe	Med-High
Rotor/blade structural failure	Likely	Severe	High
Gradual degradation	Very Likely	Minor	Low-Med
Cascading motor failures	Possible	Severe	High

4.2 Battery System Failures

In electric vertical takeoff and landing (eVTOL) aircraft, high-performance lithium-ion battery systems form the backbone of the propulsion powertrain, delivering both the energy and instantaneous power required for all phases of flight. These batteries are typically composed of hundreds to thousands of individual electrochemical cells, arranged into modules and packs, and managed by a Battery Management System (BMS) (Ellafrits, S. A., 2022). Each cell operates by storing energy chemically through reversible lithium-ion intercalation between an anode and cathode, separated by an electrolyte and porous membrane. When discharge is initiated, lithium

ions migrate from anode to cathode, producing free electrons that flow externally to power the motor drives. Because eVTOL operations involve frequent power surges during vertical takeoff, hovering, and emergency maneuvers, battery systems must be engineered for high current output, thermal resilience, and fault tolerance (Dixit, M. B., 2024). The failure of any battery subsystem, whether thermal, electrical, or mechanical, can have direct implications for flight safety and mission reliability. This section outlines the primary failure modes associated with eVTOL battery systems and mitigation strategies rooted in current research and certification best practices.

- Battery safety in eVTOLs is central to operational reliability and certification, with multiple hazards spanning thermal, electrical, mechanical, and control domains. Among these, **thermal runaway** remains the most catastrophic. Although the widely used 21700 cell format enables high energy density (>265 Wh/kg) and modularity (Dovetail, 2025), thermal abuse tests show that once initiated by internal faults such as dendrite penetration or separator failure, cell temperatures can spike above 700°C in minutes, regardless of state of charge (Shelkea, A. V., 2022). The risk compounds due to propagation—runaway in one cell can ignite adjacent cells unless interrupted. NASA and industry guidelines stress the importance of cell-level fusing and thermal barriers to halt such propagation (Darcy, E., 2016).
- **Voltage sag** under peak loads like vertical takeoff or emergency climb represents a more gradual but performance-critical risk. Empirical testing shows that as batteries age, voltage under high-discharge loads can drop by 10–20%, degrading thrust margins when they are most needed (Dixit, M., 2024; Bills, A., 2023). Adaptive power controllers and reserve margins mitigate this risk by reallocating power in real time and preserving capacity for emergency demands (He, Z., 2025).
- **Failures within the Battery Management System (BMS)** pose another serious threat. Sensor faults, such as inaccurate current or temperature readings, can lead to unsafe charge/discharge behavior or delayed cooling responses. Simulated fault-injection studies demonstrate that a single corrupt reading can cripple safety logic (Dey, S., 2016). Solutions include distributed sensing, cross-validation strategies, and sensor placement optimization to improve fault coverage and redundancy (Cheng, Y., 2022).

- **Mechanical failures**, especially under dynamic loading, also present a nontrivial risk. Drop tests of NASA-prototyped battery enclosures for eVTOLs reveal that impact forces can deform structures, damage connectors, or puncture cells (Littell, J., 2025). These findings emphasize the need for aerospace-grade vibration-rated fasteners, structural adhesives, and pressure-relief features like rupture discs and venting channels (Harwin, 2025; E-Mobility Engineering, 2023; Freudenberg, 2025).
- Finally, **internal short circuits**, stemming from latent cell manufacturing defects or operational factors like lithium plating, represent unpredictable but high-consequence events. FAA research confirms these defects may go undetected for extended periods before triggering failure (Cabler, S. J. M., 2015; Keyser, M., 2015). Rapid-response safety devices such as pyro-fuses are therefore standard in eVTOL pack design. These components disconnect faulted segments in under 2 milliseconds, preventing escalation and preserving pack integrity (Eaton, 2025).

Table 5: Risk Matrix for Battery System Failure Modes

Category	Likelihood	Impact	Risk Level
Thermal runaway	Possible	Severe	High
Voltage sag under high load	Likely	Moderate	Medium
BMS failures	Possible	Minor	Low
Mechanical failures under dynamic loading	Unlikely	Significant	Low Med
Internal Short circuits	Very Unlikely	Severe	Low Med

4.3 Power Distribution and Control Failures

The power distribution and control subsystem in an eVTOL propulsion architecture serves as the electrical and logical backbone that enables safe, synchronized thrust generation across multiple motor channels. This system includes components such as electronic speed controllers (ESCs), gate drivers, power buses, high-voltage connectors, and embedded firmware that interfaces with the flight control computer. Each element performs critical functions: managing current flow to executing precise switching sequences that regulate motor torque. Unlike conventional aerospace systems, eVTOL architectures demand high switching frequencies, ultra-fast response times, and compact layouts. This subsystem is not only responsible for performance and energy efficiency, but also plays a role in managing faults, enabling redundancy, and defending against cyber intrusion. To properly assess its failure modes and design safeguards, one must first understand the underlying electrical architecture, the dynamic interaction between hardware and control logic, and the operational stressors that drive system-level risk.

- In eVTOL propulsion systems, **Electronic Speed Controllers (ESCs)** serve as the critical link between the battery and motors, regulating power flow through high-frequency switching of power semiconductors. Their continuous exposure to high thermal loads, electrical noise, and tight packaging environments introduces multiple failure pathways. Fault analysis identifies gate signal instability and timing desynchronization as dominant degradation modes. For example, unstable or disconnected gate signals, often triggered by overheating or wire faults, can lead to component overheating and eventual ESC failure (Marcos-Andrade, 2024). Similarly, desynchronized PWM switching misaligns with rotor back-EMF, leading to mis-commutation that injects harmonics into the motor, increasing heat and reducing efficiency (Gorospe, G. E., 2020). To address these risks, system designers are adopting digital phase-locked loop (PLL) architectures, such as quadrature-PLL observers, to maintain precise timing alignment while minimizing signal latency (Saha, B., 2025).
- **Gate driver** reliability is especially crucial in high-power multirotor configurations. Fast switching of MOSFETs or SiC transistors can introduce gate oscillations or false turn-on events, which may result in shoot-through—simultaneous conduction of both high-side and low-side switches. Studies show that rapid dv/dt events generate common-mode noise that

corrupts gate signals, triggering unintended switching (Song, H., 2022; Tahir, M., 2021). To mitigate this, ESC designs increasingly employ galvanic isolation for high-side drivers, ensuring signal integrity and suppressing noise-induced faults. Low output impedance drivers and fast switching times further help reduce conduction and switching losses, preserving performance under stress (Kennedy, B., 2012).

- **Busbars and high-voltage connectors** form the backbone of eVTOL power distribution, yet arc faults in these subsystems represent a severe hazard. Mechanical degradation, loose connections, or insulation breakdowns can create high-resistance zones that initiate localized arcing. These arcs can evolve into high-energy plasma events, vaporizing conductors and propagating fire across panels and nearby electronics (Eddins, C. L., 2004). To detect and suppress arcs before escalation, advanced photodiode and thermopile sensors tuned to IR/UV signatures are used to trigger fast isolation protocols. Systems like Eaton's ARCON protection units can interrupt HV arcs in less than 2 ms, meeting IEC/TS 63107 guidelines and preserving adjacent systems (Eaton, 2025).
- **Cybersecurity** further complicates system-level safety due to the increasing reliance on networked components such as the ESCs, BMS, and flight controllers. These units frequently communicate over CAN bus protocols, which lack built-in encryption or authentication. Studies draw attention to the ease with which attackers can spoof data, issue malicious commands, or disable autopilot logic if they gain access to internal buses (EE Times, 2022; Stone, J., 2019). As such, real-time authenticated messaging is critical. Solutions like Trillium's SecureCAN add symmetric encryption and authentication layers that operate within the sub-millisecond threshold, ensuring timely and verified command and sensor data exchanges (EE Times, 2022). This approach significantly reduces the risk of injected faults or control spoofing in dense urban environments where failure margins are slim.

Table 6: Risk Matrix for Power Distribution and Control System Failure Modes

Category	Likelihood	Impact	Risk Level
Electronic Speed Controllers (ESCs)	Possible	Significant	Med-High
Gate driver	Possible	Moderate	Medium
Busbars and high-voltage connectors	Unlikely	Severe	Med-High
Cybersecurity	Very Unlikely	Significant	Med-High

4.4 Risk Profiles of Critical Flight States

The following subsection will examine key hazards associated with vertical takeoff and initial climb, hover out of ground effect, and approach and vertical landing, with a focus on failure modes that emerge from the coupling of vehicle dynamics, subsystem performance, and environmental interaction.

4.4.1 Vertical Takeoff and Initial Climb

Vertical takeoff is widely recognized as one of the highest-risk phases of an eVTOL mission because it simultaneously stresses the propulsion system, challenges aerodynamic stability, and places demands on the flight control architecture. In this regime, the aircraft must generate lift solely through powered rotors without the benefit of translational aerodynamic efficiency. These conditions amplify the sensitivity of the system to any deviation in thrust, battery performance, or control authority. Moreover, the vehicle must manage rapidly changing aerodynamic interactions as it rises through ground effect and encounters rotor interference phenomena unique to distributed electric propulsion. As a result, even minor discrepancies in power delivery or control response

can cascade into destabilizing behaviors, making the initial climb a phase in which thermal, aerodynamic, and control tolerances are especially constrained.

- Vertical takeoff is one of the most taxing phases in an eVTOL flight envelope, marked by a **convergence of peak electrical demand and complex aerodynamic interactions**. Electrically, the challenge lies in the extreme C-rate demands placed on lithium-ion battery systems during high-thrust operations like liftoff, hover, and climb. Takeoff can draw 10C to 60C discharge rates, pushing batteries toward thermal and electrical limits (Dixit, A., 2024). In an Oak Ridge National Lab (ORNL) test replicating such conditions, none of the tested high-power pouch cells lasted beyond 100 cycles; most declined noticeably after 85 (Beadle, A., 2024). However, a 20% reduction in peak power, combined with moderate depth-of-discharge and controlled temperatures, extended cell life to 2,348 cycles in simulation, demonstrating how modest power and thermal management can yield dramatic durability improvements (Bills, A., 2023). This tradeoff stresses the value of performance-aware energy management strategies for preserving battery longevity under peak load conditions.
- Aerodynamically, takeoff begins in the **in-ground-effect (IGE)**, where the rotor wake interacts with the surface below. In multi-rotor eVTOLs, this effect produces a radial wall jet that rebounds as an upward “fountain” flow, where an air column is formed by the convergence of downwash jets from adjacent rotors. This fountain flow introduces turbulence and pressure variability near the rotor plane, reducing local thrust efficiency (Zhu, H., 2024). Rotor layout can help mitigate this effect. For example, experiments show that reducing rotor spacing weakens fountain flow formation and restores near-ground thrust performance in quadrotors (Otsuka, H., 2024).
- During the **IGE-to-Out of Ground Effect (OGE) transition**, as the vehicle climbs, asymmetric ground proximity can induce uneven lift. When only some rotors remain in ground effect, such as over a sloped rooftop or raised pad, the resulting torque imbalance creates roll or pitch perturbations that standard PID controllers struggle to correct. Tests reveal that such partial ground effect can cause unexpected pitch-ups and oscillations (Cuevas, P. S., 2017). A mitigation strategy involves feed-forward control: embedding a ground-effect model into the flight controller allows the system to anticipate lift asymmetry

and apply corrective inputs preemptively. Adding this model-based term significantly improved pitch stability in experiments, suggesting that predictive compensation is more effective than feedback-only control in near-ground conditions (Cuevas, P. S., 2017).

Table 7: Risk Matrix for Propulsion Hazards During Takeoff and Transition

Category	Likelihood	Impact	Risk Level
Peak electrical / thermal stress	Likely	Significant	Med-High
Ground effect induced thrust asymmetry	Possible	Moderate	Medium
Thrust imbalance during IGE–OGE transition	Possible	Significant	Med-High

4.4.2 Hover

Hovering flight challenges eVTOL systems with sustained power demand, limited control authority, and minimal aerodynamic support. Without translational lift, any disturbance or system degradation, such as sensor dropouts or actuator delays, can immediately destabilize the aircraft. Most multirotor eVTOLs rely on rotor speed modulation, which introduces latency and reduces responsiveness to gusts or failures. This section examines the thermal, control, and redundancy constraints that make hover a critical and tightly constrained flight regime.

- Hover flight presents **critical stability and redundancy challenges** for eVTOL vehicles, especially multirotor designs with fixed-pitch rotors. Unlike helicopters that modulate lift and orientation through blade pitch (via swashplates) for immediate response, most eVTOLs rely on RPM variation to control thrust, introducing response lag. This delay

becomes problematic in gusty or dynamic environments. Studies comparing control modalities confirm that RPM-only systems act more like acceleration controllers, slow to adjust while blade pitch offers direct velocity control (Pavel, M., 2021). NASA's Assisted Hover Automation (AHA) initiative addresses this latency through predictive control augmentation. In simulation, AHA significantly reduced pilot workload and cross-track deviation in challenging 17-knot crosswinds, indicating the benefit of intelligent control overlays for hover-phase stability (Kaneshige, J., 2024).

- **Redundancy**, another pillar of safe hover operations, becomes **critically strained during out-of-ground-effect (OGE) flight**. In multirotor configurations, fault tolerance depends on available thrust margin. If a vehicle hovers near full power, the loss of a rotor leaves insufficient headroom for the remaining motors to compensate. For example, in one NASA experiment, a full-scale UAM hexacopter operating with low thrust margins could increase rotor speed by only ~6% before hitting its Maximum Continuous Power limit, rendering fault recovery impossible (Malpica, C., 2022). These findings underscore that thrust redundancy must be baked into the design, not assumed. Engineers recommend over-sizing propulsion to maintain headroom for fault recovery and optimizing control allocation logic to handle actuator saturation and maintain torque balance after failures (Malpica, C., 2023).
- Hover control also hinges on the **integrity of state estimation systems**, which fuse data from IMUs, GPS, and other sensors to track vehicle motion in real time. However, these inputs can degrade: GPS signals may be blocked or distorted in urban or low-altitude settings, and IMUs are prone to drift from integration bias (Xiang, S., 2024). These errors propagate into the flight control loop, undermining stability. To mitigate this, engineers integrate visual odometry and LiDAR-based localization to supplement traditional sensors. Vision systems track surface features to estimate motion, while down-facing LiDAR provides accurate altitude and environment mapping, enhancing resilience in GPS-denied conditions (UAV Navigation, 2022). Together, these redundant sensing pathways support robust hover performance across diverse and challenging operational environments.

Table 8: Risk Matrix for Propulsion Hazards During Approach and Vertical Landing

Category	Likelihood	Impact	Risk Level
Peak power demand with degraded batteries	Likely	Significant	High
Vortex Ring State	Unlikely	Severe	Med-High
Loss of propulsion without autorotation capability	Very Unlikely	Severe	Medium

4.4.3 Approach and Vertical Landing

The final approach and vertical landing phase of eVTOL operations concentrates multiple operational risks into a narrow time and altitude window, making it one of the most technically demanding segments of flight. As the aircraft decelerates into a hover and prepares for touchdown, power demand peaks just when energy reserves are lowest, thermal margins are thinnest, and aerodynamic conditions are most unstable. This convergence of thermal, electrical, control, and aerodynamic stressors exposes critical vulnerabilities in propulsion and flight systems. The following section addresses three primary risks associated with this phase: battery strain during peak-power demand, susceptibility to vortex ring state (VRS), and the absence of effective autorotation capability in most eVTOL designs.

- The final descent and vertical landing phase of eVTOL operations compresses several critical risks into a narrow temporal and altitude window. **Thermal, control, electrical, and aerodynamic stressors all converge** just as the vehicle decelerates to a hover, often

with partially depleted and thermally burdened batteries. This poses a challenge for pure-battery architectures, which must deliver high peak power despite degraded conditions. To address this, hybrid-electric propulsion strategies, combining fuel cells or compact generators with batteries, have emerged as a viable solution. These architectures offload cruise power demands to the non-battery source, preserving battery headroom for final-phase demands like turbulence or hover (Careless, J., 2021).

- The greatest concern in vertical landing is the **vortex ring state (VRS)**. The Vortex Ring State (VRS) is a complex aerodynamic phenomenon on the main rotor, which occurs at low air speeds, usually during the critical segments of final approach and landing. The VRS involves a rapid and substantial loss of rotor lift and control power. When VRS is not prevented (flight trajectory & power), nor detected (early signs), and in proximity to the ground, it can have serious consequences (Airbus, 2019). To mitigate VRS, pilots must limit descent rates to under 500 feet per minute below 30 knots and avoid rapid decelerations or steep approaches at low air speeds, especially in tailwinds or confined environments. In case of early VRS symptoms, immediate recovery involves adding power and applying forward cyclic to regain airspeed and exit the vortex (Airbus, 2019).
- Compounding the challenge is the **lack of effective autorotation capability** in most eVTOLs. Unlike helicopters, eVTOLs with low-inertia rotors, especially in distributed electric propulsion (DEP) designs, cannot sustain lift in a total power loss scenario. This increases the risk of uncontrolled descent from low altitude (Shubert, M., 2025). Instead of autorotation, the FAA allows “equivalent maneuver” alternatives that ensure controlled emergency landings (CELs). These include fail-operational architectures, redundant flight controls, partial-power descent paths, and real-time energy management and landing site prediction. Together, these technologies aim to achieve an equivalent level of safety (ELOS) despite the inherent limitations of eVTOL rotor design (Shubert, M., 2025).

Table 9: Risk Matrix for Propulsion Hazards During Hover

Category	Likelihood	Impact	Risk Level
Insufficient thrust margin after rotor loss	Possible	Severe	High
RPM-only thrust modulation latency	Likely	Moderate	Med-High
State estimation degradation	Possible	Moderate	Medium

5 Analysis of Risks and Safety Enhancement

Sections 1 and 2 establish the operational context and environment for UAM towards laying the groundwork for identifying propulsion-related hazards through a STPA. Section 3 applies STPA to identify system-level hazards (H-1 through H-5) and their propulsion-specific sub-hazards, revealing the control structure of interactions that can lead to unsafe outcomes. Then Section 4 combines the understanding of the operational environment along with the hazard identification from the STPA towards creating an in-depth component level analysis.

Building on this foundation, this section now quantifies and prioritizes these risks using analytical methods such as Failure Mode and Effects Analysis (FMEA) and Fault Tree Analysis (FTA) to help identify the mitigation strategies necessary to achieve acceptable safety levels of 10^{-9} . The safety objective of 10^{-9} per flight hour is derived from long-standing aircraft system safety practice, where an aircraft-level catastrophic failure rate of 10^{-7} is allocated across multiple independent catastrophic failure conditions identified through Functional Hazard Assessment. This per-failure target has become the accepted benchmark for catastrophic events in transport-category aircraft and is now being applied to passenger-carrying eVTOL operations under performance-based certification frameworks (Vincent, J., 2021).

5.1 Propulsion System Architectures

The first step in characterizing the risks of UAM propulsion systems is understanding how individual component failures can combine into more severe events. A common method for ranking risks is the Failure Mode and Effects Analysis (FMEA), which analyzes how each part might fail and assesses the consequences. Since UAM vehicles are highly complex systems with many interconnected components spanning electric motors, batteries, and power electronics, it is also necessary to consider how failures spread from one component to another. Each failure is described by how likely it is to occur (Occurrence), how severe it would be (Severity), and how easily it can be detected before causing harm (Detection). The product of these three factors yields the Risk Priority Number (RPN), which guides prioritization of mitigation efforts.

Table 9 presents the FMEA results for two critical subsystems identified in Section 3: the battery system and the tilt mechanism. The analysis reveals that tilt actuator jam (RPN=180) represents the highest-priority failure mode due to the combination of maximum severity and moderate detection difficulty. A jammed actuator during the transition phase would trap the aircraft between vertical and cruise modes, potentially leading to loss of control with minimal recovery options. BMS sensor faults (RPN=140) rank second, as incorrect temperature or state-of-charge readings can mask developing thermal issues or lead to energy exhaustion before landing. Thermal runaway (RPN=120) carries the maximum severity rating due to its potential for fire, explosion, and toxic gas release, though its lower occurrence rating reflects the multiple protective barriers typically implemented in aviation-grade battery systems.

Table 10: FMEA – Battery System and Tilt Mechanism

Component	Failure Mode	Effect	Severity (1-10)	Occurrence (1-10)	Detection (1-10)	RPN	Linked Hazard	Cause	Mitigation
Battery System									
Li-ion Cell	Thermal Runaway	Fire/explosion, loss of power, toxic gas release, potential hull loss	10	3	4	120	H-2.1, H-2.4 (L-1, L-2, L-4)	Dendrite penetration, separator failure, internal short	Cell fusing, thermal barriers, pyro-fuses, venting systems

Battery Pack	Voltage Sag (High Load)	Reduced thrust margin, inverter cutoff during go-around attempt	8	6	2	96	H-3.1, H-1.1 (L-1, L-2, L-3)	Battery aging, high C-rate discharge, low SoC	Reserve margins, adaptive controllers, SoC monitoring
BMS	Sensor Fault (Temp/SoC)	Incorrect SoC estimate, delayed thermal response, energy exhaustion	7	4	5	140	H-3.2, H-2.1 (L-1, L-2, L-3)	Sensor drift, wiring fault, EMI corruption	Redundant sensors, cross-validation, HMS integration
Tilt Mechanism (Vectored Thrust Configurations)									
Tilt Actuator	Mechanical Jam/Seize	Aircraft trapped in vertical or cruise mode, loss of control during transition	10	3	6	180	H-1.3, H-1.1 (L-1, L-2, L-5)	Mechanical wear, debris ingestion, bearing failure	Redundant actuators, position monitoring, safe-mode logic
Position Sensor	Erroneous Reading	FCC commands incorrect thrust vector, asymmetric lift, attitude upset	8	4	3	96	H-1.2, H-1.3 (L-1, L-2)	EMI, vibration, connector loosening, calibration drift	Dual sensors, cross-validation, back-EMF estimation

The transition between vertical lift and wing-borne flight represents the most vulnerable phase for vectored-thrust and lift-plus-cruise configurations. During this maneuver, the aircraft operates in a complex aerodynamic state with changing thrust vectors and high angles of attack (May et al., 2025). A failure during transition is particularly dangerous because the aircraft may have insufficient forward airspeed for wing-borne flight while simultaneously lacking the rotor authority for stable hover.

Figure 9 presents the FTA for H-1.3 (Loss of Control During Transition), decomposing the top event into three primary failure branches: tilt mechanism failure, propulsion system failure, and control system failure. Each branch connects through OR gates to basic events representing specific component failures. The propulsion branch includes motor failure, ESC failure, and battery voltage sag with the latter representing the most likely contributor given that transition often occurs when batteries have already been partially depleted by the power-intensive takeoff and climb phases. The AND gate at the bottom of the tree captures cascading scenarios where primary motor loss leads to overload-induced failure of remaining units.

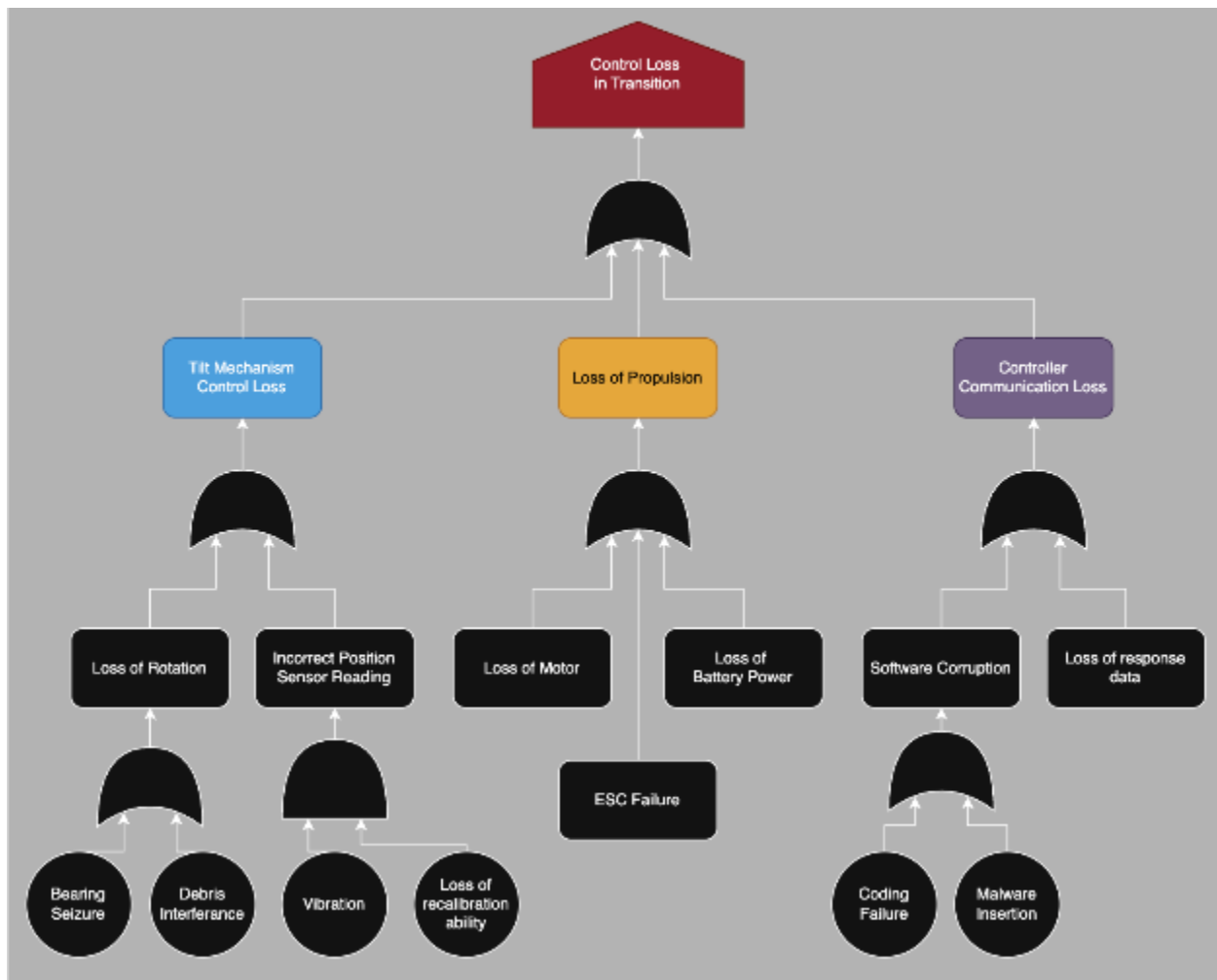


Figure 9: Fault Tree Analysis for Loss of Control During Transition Phase

The landing phase presents similarly elevated risk levels due to the convergence of multiple stress factors: power demand peaks again for hover and descent arrest, batteries are at their lowest state-of-charge, and thermal margins have been eroded by the preceding flight. This combination makes

the final approach particularly vulnerable to voltage collapse if batteries cannot sustain the instantaneous power required for emergency maneuvers such as go-arounds (Dixit, 2024).

5.2 Mitigation Strategies

Addressing the risks identified through FMEA and FTA requires a defense-in-depth approach spanning hardware, software, and operational domains.

Hardware Mitigations: Redundancy through independent motor and battery channels prevents single points of failure, though this redundancy must extend to power distribution and control pathways to be effective. Thermal management via liquid cooling loops and predictive temperature monitoring maintains components within safe operating limits during high-power phases. Pyro-fuses and thermal barriers can contain battery failures within milliseconds, preventing cell-to-cell propagation (Eaton, 2025). For tilt mechanisms, redundant actuators and continuous position monitoring enable detection and accommodation of mechanical failures.

Software Mitigations: Fault-tolerant control algorithms can redistribute thrust among remaining motors within milliseconds of detecting a failure, maintaining aircraft stability during partial-power operation (Su, 2024). Health monitoring systems cross-validate sensor data from multiple sources to detect incipient failures before they cascade. Adaptive power controllers preserve energy reserves for emergency demands by reallocating loads in real time.

Operational Mitigations: Pre-flight automated diagnostics verify all systems before each departure. Conservative energy planning ensures adequate reserves for go-around maneuvers and accounts for battery degradation over the aircraft's service life. Vertiport designs incorporate fire-resistant surfaces and safe standoff distances to protect ground personnel.

5.2.1 Integrated Safety Framework

Achieving the 10^{-9} fatal accident rate target established by the FAA and EASA requires these multiple layers of protection working together as an integrated system (NASA, 2021). No single mitigation is sufficient; safety emerges from the combination of robust hardware design, intelligent software responses, and conservative operational procedures. Digital twin technology offers a promising path forward by enabling real-time comparison between actual and predicted system behavior, supporting predictive maintenance and early anomaly detection. The STPA allows for pre-emptive hazard identification whereas the FMEA and FTA analyses presented in this section

provide the qualitative/ quantitative foundation for prioritizing design improvements and demonstrating compliance with emerging certification standards for UAM aircraft.

6 Conclusion

Urban Air Mobility is much different than traditional aviation because it involves a large number of electric aircraft flying at low altitudes in densely populated environments. These aircraft fly close to buildings and people, meaning there is little time or space to recover in the event of emergency. This project focused on understanding the risks that come from propulsion system failures during the most critical phases of flight; takeoff, hover, transition, and landing.

By looking at propulsion systems using both system-level (STPA) and component-level (FMECA/ FTA) analysis, this study showed that failures in eVTOL aircraft are almost never isolated events. They are typically problems that spread from one part of the system to another. In DEP systems, a failure in one motor can lead to excess stress on the other motors, batteries, and controllers. These chain reactions can happen very fast, typically quicker than a pilot or operator could react, which makes automated fault detection and response very important.

The analysis showed that the level of risk changes depending on the flight phase. Terminal operations like takeoff and landing are dangerous because they require high power while the aircraft is close to the ground. During these phases, batteries are under heavy load and thermal limits are pushed, leaving little to no room for error. External factors like wind, debris, or electromagnetic interference can also make these phases even more difficult to manage and mitigate failures.

To reduce these risks, multiple layers of protection are necessary. Starting with strong hardware designs, like redundant motors, thermal protection, and electrical isolation, can help prevent failures. The next important layer is in software design solutions. Incorporating fault-tolerant control systems and health monitoring can help detect problems early and keep the aircraft stable if a failure occurs. Finally, operational and certification measures, like maintenance, safe vertiport design and conservative energy planning also play a role in protecting both passengers and people on the ground.

Overall, this project shows that making UAM safe is not just about preventing individual components from failing. It is about understanding how failures interact across the entire system and during different flight phases. Future work should focus on better modeling of these failure chains and developing safety strategies that account for how quickly risks can grow in real-world urban operations. Furthermore, safety goes beyond engineering; it requires addressing the human element. Future research must also investigate the human psychology of machine-human teaming to build trust in high-level automation. Specifically, the industry must develop strategies to address the general population's "automation phobia" regarding pilotless aircraft operating over their homes, ensuring that the public perceives these systems not just as efficient, but as inherently safe.

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